Skills Summary

Conditional and Independent-Event Probabilities

Use this reference while completing your worksheet or when studying.

Skill 1 — Conditional probability from counts

Definition: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{\text{count in both}}{\text{count in } B}$.

Conditioning *shrinks the sample space* to B: in a two-way table, work inside B's row or column and divide by that row or column total, never by the grand total. $P(A \mid B)$ and $P(B \mid A)$ are different questions.

Example: Of 50 club members, 30 play chess, and 18 of those chess players also play go. Find $P(\text{go} \mid \text{chess})$.
Step 1. The condition is "plays chess", so the sample space shrinks to those 30 members and 50 stops being the denominator.
Step 2. $18 \div 30 = \mathbf{\dfrac{3}{5}}$, so about 60% of the chess players also play go.

Try it: Of the 40 students taking Spanish, 12 also take art. Find $P(\text{art} \mid \text{Spanish})$.
check: $\mathbf{\dfrac{3}{10}}$

Skill 2 — The multiplication rule

General rule: $P(A \cap B) = P(A) \cdot P(B \mid A)$.

Drawing **without replacement** makes the second factor conditional: both the favourable count and the total drop by one. Only when the events are independent does the second factor stay equal to $P(B)$.

Example: A bag holds 10 marbles, 4 of them blue. Two are drawn without replacement. Find $P(\text{both blue})$.
Step 1. The draws are not independent, so use the general rule with a conditional second factor.
Step 2. $\dfrac{4}{10} \times \dfrac{3}{9} = \dfrac{12}{90} = \mathbf{\dfrac{2}{15}}$.

Try it: A deck of 15 cards holds 6 red cards. Two are drawn without replacement. Find $P(\text{both red})$.
check: $\mathbf{\dfrac{1}{7}}$

Skill 3 — Testing independence

Test: A and B are independent exactly when $P(A \cap B) = P(A) \cdot P(B)$.

Equivalently, $P(A \mid B) = P(A)$: conditioning on B changes nothing. Compute the product, compare it with the given intersection, and say which it is — independence is a conclusion you check, never an assumption you make.

Example: $P(A) = 0.5$, $P(B) = 0.3$, $P(A \cap B) = 0.15$. Independent?
Step 1. Multiply the two marginal probabilities and compare the product with the given intersection.
Step 2. $0.5 \times 0.3 = \mathbf{0.15}$, which equals $P(A \cap B)$, so A and B are independent.

Try it: $P(A) = 0.7$, $P(B) = 0.2$, $P(A \cap B) = 0.1$. Compute $P(A) \cdot P(B)$ and decide.

check: **0.14**, not 0.1 — dependent

Skill 4 — Independent trials and the binomial term

Three independent trials: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, where $a = P(\text{success})$ and $b = P(\text{failure})$.

Each term is one outcome count: $3a^2b$ is exactly two successes, and the coefficient 3 counts the orders in which the single failure can fall. Independence is what lets the probabilities in a term be multiplied.

Example: Three independent shots, each made with probability 0.7. Find the probability of exactly 2 makes.
Step 1. Exactly two successes is the $3a^2b$ term, so substitute $a = 0.7$ and $b = 0.3$ into it.
Step 2. $3 \times 0.7^2 \times 0.3 = 3 \times 0.49 \times 0.3 = \mathbf{0.441}$.

Try it: Three independent shots, each made with probability 0.5. Find the probability of exactly 2 makes.

check: **0.375**